Readout Universe

Philosophy and Logic

An Open-Science, Reproducible Textbook

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Open Science and Reproducibility Statement

This textbook is built on a single non-negotiable rule inherited from its parent research programme: *deliver operationally before claiming symbolically* (T_{op} before T_{sym}). Every load-bearing claim in this book therefore falls into exactly one of the following classes, and says so in the margin of the claim itself:

1. [Th_coqc] — machine-checked in Coq (v8.18), axiom-free over the discrete core: the proof script is reprinted in Appendix B, compiles with `coqc` exit code 0, and every theorem answers `Print Assumptions` with *Closed under the global context.*
2. [finite_diagnostic] — verified by an executable Logic Test Protocol (LTP): the exact Python source is reprinted in Appendix A; running it reproduces the quoted numbers. A finite diagnostic is *evidence*, never a proof.
3. [Dr] / [Th*] — derived under an explicitly declared bridge assumption (each bridge is named, numbered, and listed in Appendix D).
4. [Cj] / [Wf] / [Open] — conjecture, working form, or open problem. An [Open] never appears bare: the author's current stance is always attached.

To reproduce everything in this book:

```
# in the bundle root (Python 3 + numpy; Coq 8.18 for the formal floor)
python3 code/LTP1_logic_as_residual_flow.py    # expect: PASS 3/3
python3 code/LTP2_3_4_battery.py               # expect: PASS 8/8
coqc code/UPL_Sorites.v                        # expect: exit 0 and three times
                                               # "Closed under the global context"
```

A claims register with one falsifiable statement per row and its exact check command is given in Appendix C. If you find a genuine falsification, you have not embarrassed this book; you have used it exactly as designed.

AI assistance disclosure. The research direction, the core ideas, and all final decisions are the work of the human author. AI assistants were used in an assistant role only — formal-proof drafting to the author's specification, numerical implementation, adversarial self-review, and editorial structuring.

Lineage. This volume is the English, textbook-form edition of the UPL programme (Universal Philosophy & Logic), a lens over the author's physics–philosophy canon *Genesis of the Universe, Step by Step* and its machine-checked successor system. The parent canon retains ontological authority: where this book and the canon disagree, the canon wins.

How to Read This Book: The Reading Contract

1. **One language.** Everything in this book is causal-informational structure on a finite discrete graph. Physics, logic, mind, and society are *one stream read at different causal-memory times* τ_c. Nothing is "translated between domains"; we only slide along the stream.

2. **Readout, not truth.** Everything a finite agency reads is a finite, discrete *readout*. The continuum, $\mathbb{R}$, and actual infinity are *non-readouts*. Where a proof needs them, the tier tag says so.

3. **The book is a testable engine.** Each chapter ends with the condition under which that chapter is wrong. The universal falsifier of the whole programme: exhibit any observer with $\varepsilon_{\text{tot}} = 0$ (a system whose representation of the world coincides with the world) and every page of this book collapses.

4. **No silent lifts.** Every strengthening of a claim between versions is recorded in the regression log (Appendix D), including the retractions.

Contents

Part I

The Root

Chapter 1

The Third Stance: Logic Is Neither Eternal Nor Conventional

Western philosophy of logic has long oscillated between two poles. The first is *logical Platonism*: the laws of logic are eternal truths, prior to the world and to any thinker (Frege carried this banner). The second is *conventionalism or psychologism*: logic is a human agreement, or a habit of the brain — the pole Frege and Husserl attacked so effectively that it nearly vanished from twentieth-century philosophy.

This book proposes a third stance, parallel to the informationalism of the parent canon:

Principle 1.1 (The Third Stance)**.** [Dr] Logic is the structure that necessarily emerges from the condition of a *finite knower on a discrete graph.* It is the admissible transport grammar that any finite system must possess in order to bring retained distinguishabilities into comparison at all.

Against Platonism: there is no "third realm" of logical law — there is one stream of causally ordered difference, and logic is the reading of that stream at the knower's τ_c. Against psychologism: the derivation does not start from empirical human psychology but from *structural* finitude — $\varepsilon_{\text{tot}} > 0$, bounded bandwidth, $\tau_c > 0$ — constraints that bind *any* finite system, human, animal, or machine. This answers Frege's classical objection: we do not reduce logic to how humans happen to think; we elevate it to how any finite system *must* handle difference.

Falsifier 1.2. Exhibit a system that performs contentful comparison with no admissible grammar whatsoever. Then the third stance fails (cf. falsifiers U2/U3, Appendix C).

Chapter 2

Why Logic Must Have a Genesis

A standard logic textbook begins "let p be a proposition" — as if propositions, negation, and implication fell from the sky together. The question such books never ask is: *what must already exist for there to be a proposition at all?*

The parent canon answered this style of question for physics (mass is not substance but inverse memory time; time is not a container but the total sequence of asymmetric transitions). This book does the same for logic:

Principle 2.1 (Birth Certificates)**.** [Df]Every logical constant must exhibit its own birth certificate: the step of the 42-step genesis stream from which it emerges, and every bridge assumption it carries.

The principle has teeth because it can *reject*: if a logical constant requires actual infinity, direct access to the latent world θ, or $\varepsilon_{\text{tot}} = 0$, then it is at best an idealized projection, not native logic of a finite knower. Chapter 7 shows that classical bivalence falls exactly into this class.

Chapter 3

The Primordial Logic Stream (L-01 to L-04)

Axiom 3.1 (L-01: Identity and difference are born as a pair)**.** [Ax]From the primordial axiom $\exists\, a, b :\; a \neq b$, the statement $a = a$ is definable only against the background of $a \neq b$. Identity does not precede difference; the two arise together as the first contrast.

Here we stand against Leibniz (identity of indiscernibles as a first principle) and closer to Heraclitus and Nāgārjuna (Chapters 16–17).

Theorem 3.2 (L-02: Negation as exclusion)**.** [Th] *Given distinguishability (the first bit), "not this" becomes sayable: $\neg$ is definable as* exclusion within a field of alternatives *A_D.*

This is by birth a *choice negation*: "not a" means "some other member of A_D", not the absolute complement of a universe. Consequence: the law of excluded middle is licensed only when A_D has been exhaustively enumerated — and for a finite knower whose salient field satisfies $S_A \subsetneq \Delta$, exhaustiveness is never guaranteed. This is the root of our sympathy with intuitionism (Chapter 21) without Brouwer's mentalism.

Theorem 3.3 (L-03: Directed implication)**.** [Th]*From causal asymmetry ($A \to B \neq B \to A$), implication is a* directed walk along causal chains*, not a static truth table. Material implication ($p \to q \equiv \neg p \vee q$) is the projection of directed implication onto a static table; the notorious "paradoxes of material implication" are the price of that projection, not deep truths about implication.* [Dr]

Theorem 3.4 (L-04: Inference as descent)**.** [Dr] + [finite_diagnostic] *Inference is a countable path of Lyapunov descent toward consistency: $\dot{\varepsilon} = -\Lambda A^{\top} W r$, where a proof is a path on the graph.* ***Operational evidence (LTP1-C1):*** *under constant evidence, $V(0) = 0.654$ descends monotonically (zero violations) to 8.2×10^{-26}.*

This is the first time in the programme that "inference" has an executable mathematical meaning rather than a metaphor. Its precise status — a *chosen reading* supported by a finite diagnostic, with alternatives not excluded — is recorded in the Arrow Ledger (Appendix D).

Part II

The Logic of the Finite Knower

Chapter 4

The Engine: $A\varepsilon = \delta$ Read at the Knower's τ_c

The formal heart of the book is a *re-reading* (not a re-derivation) of the seed algebra of the parent programme's minimal viable foundation:

Symbol	In physics (MVF)	In this book (knower's τ_c)
A	relational Jacobian	logical grammar (admissible transport)
ε	field deviation	belief-state deviation of a finite knower
δ	observational mismatch	world-resistance readout (never θ itself)
$r = A\varepsilon - \delta$	residual	logical incoherence (epistemic residual)
$V = \frac{1}{2}r^\top W r$	Lyapunov energy	incoherence energy of a belief state
$\dot{\varepsilon} = -\Lambda A^\top W r$	dissipative flow	inference as descent

This is the parent canon's scale principle at work: *one equation, every scale; only τ_c changes.* Physics and logic are not two sciences awaiting unification; they are one stream read at two positions.

Definition 4.1 (Bridge B1 — declared openly)**.** [Dr] The belief dynamics of a finite knower instantiate residual flow in *some* admissible triple (A, W, Λ). This book claims the *existence* of the reading, not the *uniqueness* of the implementation. Uniqueness across dynamical classes is [Open] (stance: we expect a *negative* settlement — logical implementation is plural; an adjudication campaign in the style of the parent programme's M-term audit is queued).

Chapter 5

The Defeasibility Theorem (L-06)

Theorem 5.1 (Defeasibility)**.** [Th*] *(under B1) +* [finite_diagnostic] *The logic of a finite knower must be defeasible: every settled conclusion structurally carries the status "may require revision".*

Proof sketch. $\varepsilon_{\text{tot}} > 0$ (the parent programme's irreducible error floor) implies the readout δ never coincides with the latent world θ; hence for every settled conclusion there exists a trajectory of $\delta(t)$ under which that conclusion must revise. Non-monotonicity is therefore not a defect of an inference system but the *normal dynamics* of any system tracking a world it cannot access directly. □

Operational evidence (LTP1-C2): a conclusion settled at $+1.000$ revises to -1.000 when the evidence flips at $t = 30$; V spikes to 1.876 and re-descends to 2.9×10^{-13}. The "change of mind" of a logical system is captured as pure dynamics, with no ad-hoc revision machinery added (Chapter 21 compares AGM).

Remark 5.2 (Hume relocated)*.* [Dr]The problem of induction transforms: "what justifies inference from past to future?" becomes "under what bound does a tracking system operate?" The answer is input-to-state stability: bounded $\dot{\delta}$ implies bounded r. We do not *solve* Hume; we *relocate* the problem from justification to bounds.

Chapter 6

Truth as Tracking (L-07)

The three classical camps — correspondence, coherence, pragmatism — are shown to be projections of a single object:

Definition 6.1 (Truth as maintained alignment)**.** [Dr] Truth is the *event* of alignment under resistance, maintained rather than possessed. Formally: the ISS bound, bounded $\dot{\delta} \Rightarrow$ bounded r.

The correspondence face is the resistance of θ through δ (world-side); the coherence face is the descent of V (system-side); the pragmatist face is that alignment must keep *working through time*, without a terminus.
Operational evidence (LTP1-C3): under persistent noise the time-averaged incoherence $\bar{V} = 0.008 > 0$ forever: there is no state "perfectly true, now stop"; there is only tracking that is good enough. This is the operational face of $\varepsilon_{\mathrm{tot}} > 0$.

Remark 6.2 (Honesty reclassification)*.* The Arrow Ledger (Appendix D) records that this identification is a *chosen weld*: the ISS mathematics is standard control theory, re-labelled. The contribution is the reading, not the mathematics. Earlier drafts carried a derivational flavour; that is retracted (R3).

Chapter 7

Classical Logic as an Idealized Projection (L-09)

First-order logic with bivalence and monotonicity is the *ideal limit*: $\dot{\delta} = 0$ (a frozen world), $\eta = 0$ (a noiseless readout), $t \to \infty$ (unbounded inference time), whence $r \to 0$ exactly and every proposition is crisply decidable.

Proposition 7.1 (Structural isomorphism)**.** [Dr] *The relation of classical logic to native logic is isomorphic to the relation of the continuum to the discrete graph, and of overdamped classical physics ($k < k_c$) to the full dispersion relation, in the parent canon.*

Classical logic is therefore not *wrong* — it is extremely accurate in near-limit regimes (pure mathematics, where δ is pinned by definition, is the most near-limit regime there is, which is why FOL flourishes there). It fails when applied to the live world ($\eta > 0$, $\dot{\delta} \neq 0$): the problem of induction, the paradoxes of material implication, and the failures of monotonic reasoning in common sense are all symptoms of using a projection outside its regime.

Chapter 8

The Relocation of Paradox (L-10)

We apply to classical paradoxes the same discipline the parent canon applies to the hard problem of consciousness: *relocate precisely; never claim to have solved.*

8.1 Sorites: the cut is in the readout

World-side, there is only a smooth graded quantity $g(n)$; the knower reads it through a retention threshold Π as a binary predicate $H(n) = [\,g(n) \geq \Pi\,]$.

Theorem 8.1 (Sorites relocation — machine-checked core)**.** [Th_coqc] *For monotone* $g : \mathbb{N} \to \mathbb{N}$ *and threshold readout* H*:* (T1) *the readout never flips back;* (T2) *at most one flip point exists;* (T3) *the "paradoxical step"* $H(n) \neq H(n{+}1)$ *occurs exactly at the flip point;* (T4) *raising the threshold can only delay truth. All four are proved in Coq 8.18, axiom-free (`Print Assumptions`:* Closed under the global context*); the script is Appendix B.*

Operational instantiation (LTP2, [finite_diagnostic]**):** with $g(n) = 1 - e^{-n/1500}$ (smooth: $\max|\Delta g| = 6.7 \times 10^{-4}$), the binary readout has exactly one forced jump at $n = 1039$ for $\Pi = 0.5$; moving Π to 0.8 moves the jump to $n = 2414$ while g is untouched.

Conclusion. The sorites cut is a property of the Π-readout, provably not of the graded world — the price paid when a bounded-bandwidth system compresses a graded residual into a binary readout. **Honest scope:** the Coq theorem covers monotone g (exactly LTP2's regime); non-monotone g admits multiple flips.

8.2 The Liar: a residual loop without a world anchor

"This sentence is false" forces the model M_A to evaluate its own alignment with itself: its δ is its own output — a residual loop with no world-side θ to anchor it (the pure-hallucination regime of the parent canon's world-resistance axiom). [Wf] — formalization is queued as LTP5; stance: we expect a no-fixed-point/oscillation result formalizable at the Coq level.

8.3 The infinity-injection diagnosis (operator Ω_∞)

Imported from the successor system's sharpest stance: everything an agency reads is a finite discrete readout; $\mathbb{R}$ and actual infinity are non-readouts. When a paradox involves infinity, first ask *which infinity was injected*: **I1** $\mathbb{R}$-completeness; **I2** infinite divisibility ($h \to 0$); **I3** infinite scale; **I4** actual $+\infty$.

Paradox	Injection	Diagnosis (all [Dr] unless noted)
Zeno (dichotomy/Achilles)	I2	no supertask exists on a discrete stepper ($\Delta\theta > 0$)
Thomson's lamp / supertasks	I2+I4	"the state after infinitely many steps" is a non-readout
Hilbert's Hotel	I4	the strangeness is a property of a non-readout
Continuum-framed sorites	I1	discrete core closed by Theorem 8.1
Tarski hierarchy regress	I4	[Cj] — requires real work; queued

Chapter 9

Incompleteness as Identifiability (L-11)

Proposition 9.1 (True but unreachable, constructed)**.** [finite_diagnostic] *Take the grammar A of LTP1 (rank $3 < n = 4$). Two world configurations $\theta_1 \neq \theta_2$ differing along the null space ($\|\theta_1 - \theta_2\| = 2.6$) produce* identical *records $A\theta_1 \equiv A\theta_2$; every inference trajectory of the knower coincides forever, and the Fisher matrix is singular (rank $3/4$, $\det = 0$). There exist truths that are real and permanently inaccessible from any record — as an* identifiability statement*, not mysticism.*

Declared limit. This is a *structural analogy* to Gödel: we have not constructed Gödel's self-referential sentence. Elevating the analogy to a formal bridge is [Open] (stance: we expect a mapping through the null space of a self-referential grammar — not done, not claimed). Claiming "this book solved Gödel" is an auto-fail of its own claim discipline.

Part III

Philosophy Re-read

Chapter 10

Ontology (Stream Steps 01–15)

"What exists?" transforms into "which eigenmodes have not yet decayed?" Three strongest consequences: (1) *substance ontology falls* — there is no "thing" beneath properties, only temporarily retained patterns of difference; mass itself is inverse memory, not stuff. (2) *Impermanence is a theorem*, not a doctrine: every mode decays, $|a_k[n]| \leq |a_k[0]|\, e^{-\gamma_k n \Delta\theta}$. (3) Several classical ontological questions become ill-formed — "what was before the Big Bang?" commits the same grammatical error as "which direction is north of the North Pole?" — and the method operator Ω_{phil} (Chapter 22) has a grammar gate built to catch exactly this class.

Chapter 11

Epistemology (Steps 18–28)

Post-Gettier epistemology circles "justified true belief plus what?". This book changes the unit of analysis: knowledge is not a *state* to be audited for three properties but an *event* of alignment under resistance. Justification becomes legitimacy conditions: a knowledge claim is legitimate when (a) real world-resistance is in the loop (no hallucination), (b) the correction channel is open, and (c) the claim does not exceed its accessibility class. Foundationalism, coherentism, and reliabilism become three projections of one condition set — exactly as the three truth theories were unified in Chapter 6. [Dr]

Remark 11.1 (Gettier re-read). [Dr]A Gettier case is a situation where r is low *by accident*: alignment occurs but is not maintained by genuine coupling to θ. Compare LTP1-C2: a genuinely tracking system must re-descend when δ changes; a system that was "right by luck" does not respond. Knowledge differs from true belief in its *dynamics*, not by one more static property.

Chapter 12

Philosophy of Mind (Steps 29–34)

Direct inheritance from the parent canon: the knower forms *inside* the epistemic horizon, out of error — it does not pre-exist (step 29); phenomenalization requires embodiment ($\Xi_A > 0$); and the hard problem is *relocated* — to the gap between the necessary conditions and phenomenal character — never claimed solved (step 34). That discipline is this book's model everywhere: relocating a problem to sharper coordinates is worth more than a hollow declaration of victory. For AI, the parent canon has already fixed the stance: an AI is not a knower ($\Xi_A = 0$) but an accelerator of knowing — a sentence under which this book itself was produced.

Chapter 13

Philosophy of Science (The Operator Calculus)

Popper gave falsifiability as a demarcation line but no *calculus* of where to repair after a refutation; Kuhn gave paradigm shifts but no formal mechanism. This book supplies both: methodology is an executable operator loop, and refutation has *coordinates* — each falsifier points at the step it breaks. Kuhn's "crisis" is the state in which residuals accumulate while the identifiability gate says parameter repair is insufficient: the grammar A itself must change. Paradigm = A; normal science = descent under fixed A; revolution = change of A. [Dr]

Principle 13.1 (Readout-vs-readout falsification)**.** [Dr] A refutation benchmarked against an infinity-injected intermediate is invalid. A valid test must compare readout against readout — predict the finite measured quantity directly. This is one notch sharper than bare Popperianism: it filters *false refutations*, not only false theories.

The discipline "identifiability before claiming" (orbit data identify GM, never G and M separately) is likewise elevated here from good practice to a binding rule of method — something mainstream philosophy of science still lacks as an enforced gate.

Chapter 14

Ethics (L-16)

What is derivable [Dr]: with several finite knowers, each with $\varepsilon_{\text{tot}} > 0$ and different accessibility, no one holds θ; shared correction is therefore a *structurally forced* feature of co-existence without systemic collapse. Its conditions — open channels, the prohibition on destroying one another's repair capacity, the obligation to hear readouts one cannot access oneself — are the structural core of ethics.

What is not derivable (declared): first-order moral content. This book provides the frame of legitimate deliberation, not ready-made verdicts; claiming more is an auto-fail.

Remark 14.1. [Cj]The structure lands closer to discourse ethics (legitimate conditions of deliberation) than to utilitarianism or deontology — but it is obtained from the ontology of error, not from a theory of communication. The detailed comparison is open work.

Chapter 15

Aesthetics and Political Philosophy (Open Frontier)

Both fields lack a hard derivation chain in version 1.0 and are therefore tagged [Cj] throughout. *Aesthetic experience* may be read as the perception of excellent compression — a pattern that reduces V greatly at small description length (cf. elegance in mathematics); no LTP exists yet. *Politics* reads as institutions qua shared-correction infrastructure at scale; the legitimacy of an institution is its preservation of correction channels, not its being right. This chapter is short by design: a world-class textbook is not one that covers everything, but one that never lies about what it covers.

Part IV

Dialogue with World Traditions

Discipline of this part: every comparison is interpretive ([Df]/[Dr]*). We state agreement, difference, and what this framework adds. We claim no supersession, and we refuse anachronism — ancient traditions are not claimed to have "already known" these results. Every analogy row carries an explicit mapping and is registered; refuted analogies are registered too (the false-equivalence immune system, Appendix D).*

Chapter 16

Greece: Heraclitus to Aristotle

Heraclitus — *panta rhei* plus *logos*: the closest Greek to this book. The world is flux with an *order of change* (logos $\approx$ causal ordering of difference). What we add: the logos becomes one runnable equation. **Parmenides** — the exact opposite pole: one unchanging Being is the choice of identity before difference, which axiom L-01 rejects. **Aristotle** — the Organon read as the first systematic codification of the classical projection (Chapter 7): genuinely great, and correct within its regime. Deep difference: Aristotle takes essence/substance as primary (rejected in Chapter 10); yet his *phronesis* and the defeasible reasoning of the *Topics* are closer to the Defeasibility Theorem than standard textbooks admit. [Df]

Chapter 17

India: Nyāya and Buddhist Logic

Nyāya — *pramāṇa* theory (legitimate channels of knowledge) is a pre-modern legitimacy-conditions theory: perception, inference, comparison, and testimony each carry their own validity conditions, mapping strikingly onto accessibility classes. **Dignāga and Dharmakīrti** — *apoha* (meaning as exclusion) corresponds precisely to Theorem 3.2, negation as exclusion within a field of alternatives; the *trairūpya* (three conditions of a valid reason) is a theory of admissible transport for inference. **Nāgārjuna** — śūnyatā read as isomorphic to "no substance, only relational eigenmodes" (Chapter 10), and the *catuṣkoṭi* as the rejection of bivalence outside its regime — consistent with Chapter 7 without reading Nāgārjuna as an irrationalist. The parent canon's declared Kant–Buddhist bridge ($\varepsilon_{\text{tot}} = \varepsilon_{\text{universal}} + \varepsilon_{\text{conditioned}}$) is unfolded across this part. [Df]/[Dr]

Registered refusal (immune system): "catuṣkoṭi = Priest's dialetheism" is *refuted* in our reading — Nāgārjuna rejects bivalence outside its regime; he does not assert true contradictions.

Chapter 18

China: The Mohist Canons

The Mohist Canons develop a theory of names and kinds, of reasons, and of disputation, distinctive for being more pragmatic–operational than the Greek line: the three tests judge a claim by its root, its evidence, and its use — closer to this book's "deliver before claiming" than to Greek apodicticity. What we add: a formal engine under that operational instinct. Gongsun Long's "a white horse is not a horse" reads through this framework as a grammar-level fault: comparison across categories without a declared transport. [Df]

Chapter 19

Islamic Logic: Avicenna

Avicenna separates essence from existence and develops a temporal–modal syllogistic finer than Aristotle's; binding modality to *time* (necessary = at all times, possible = at some time) resonates with this programme's reading of modality through modal difference (chance is not first; the retained field of alternatives is). Difference: his Necessary Existent is a metaphysical closure this framework does not accept — our system closes at primordial difference, not at necessary being. [Df]

Chapter 20

Kant to Frege

Kant — the closest modern relative: phenomena/noumena maps to record/world (M_A/θ); conditions of possibility map to $\varepsilon_{\text{universal}}$. Two deep differences: (1) the categories are not fixed a priori — the grammar A is revisable (paradigm change, Chapter 13); (2) we supply the *dynamics* of revision, which Kant lacks. **Hegel** — dialectic reads as residual dynamics in narrative form: thesis–antithesis as high r, synthesis as descent to lower V under an enlarged A; but Hegel claims a terminal closure (the Absolute) that $\varepsilon_{\text{tot}} > 0$ forbids. [Cj] **Frege** — father of FOL and enemy of psychologism: we accept the Begriffsschrift's achievement fully *as the calculus of the limit* (Chapter 7), and Chapter 1 answers his anti-psychologism: derivation from structural finitude is not psychologism.

Chapter 21

The Twentieth Century to Now

Gödel/Tarski — Chapter 9 gives the identifiability reading with declared limits; Tarski's hierarchy reads as the prohibition of residual loops without a world anchor (the liar, Chapter 8). **Brouwer/intuitionism** — we agree excluded middle is not universal law, but obtain this from the ontology of finite access, not from mentalism; consequence: constructive logic is the native logic of the regime $S_A \subsetneq \Delta$. [Dr] **Quine** — the web of belief with no statement immune matches the revisability of all of A; but Quine has no engine — we supply V, the flow, and the ISS bound. **AGM belief revision** — the closest formal neighbour: AGM gives static postulates of revision; LTP1-C2 exhibits revision as *dynamics* with no added postulates. The trade: AGM has representation theorems this programme does not yet match ([Open]; stance: LTP6, a postulate-by-postulate check, is the queued first step). **Non-monotonic logic** (Reiter, McCarthy) — this book is non-monotonic by birth (the Defeasibility Theorem). **Paraconsistency** (Priest) — we do not accept true contradictions in the world; contradiction lives in the readout (LTP2 is the prototype: the contradiction appears at projection, never in g). [Dr] **Peirce** — fallibilism and truth as the limit of inquiry sit close to Chapter 6; difference: Peirce puts the limit at the end of infinite inquiry; we show even the limit is unreachable ($\bar{V} > 0$ forever, LTP1-C3) — our fallibilism is stronger and quantitatively bounded.

Part V

Method and Discipline

Chapter 22

The Operator Ω_{phil}: Three Worked Problems

The solve loop: (1) **encode** the problem as a difference-question on the stream; (2) **grammar gate** — is it well-formed in this ontology? (now including the infinity-injection check Ω_∞); (3) **locate** the step and τ_c; (4) **bridge audit** — every bridge assumption downgrades [Th] to [Dr]; (5) **residual form** where possible; (6) **answer with tag**; (7) **state the falsifier of the answer itself**.

22.1 Is there free will?

Grammar gate: if "free" means exempt from causal chains, the question is ill-formed (everything lives on the causal graph). If "free" means agency — choice under constraint that preserves repair capacity — it is well-formed, located at the agency step. Answer [Dr]: libertarian free will, ill-formed; structural-compatibilist free will, real as repair-preserving agency. Falsifier: exhibit agency functionally identical in every respect with no internal model load-bearing in the loop.

22.2 Is mathematics discovered or invented?

The grammar gate catches a false dilemma: on the stream the answer is *both, at different layers*. Difference-structure (e.g. spectral truths of graphs) is *discovered* (world-side, resistant); symbol systems and idealized limits (FOL, the continuum) are *invented* as good projections (knower-side). [Dr] Falsifier: exhibit a mathematical truth with neither world-resistance nor projection status.

22.3 Why is there something rather than nothing?

Grammar gate: identifying "absolute nothing" requires distinguishing it from "something" — contentful nothing presupposes difference; the very form "why X rather than Y" presupposes a field of alternatives, which is modal difference, which is already something. The question is *ill-formed in its absolute form*: it eats its own tail. [Dr] This is what the primordial axiom gives for free: difference needs no explanation from something deeper, because any explaining already uses difference.

Chapter 23

The LTP Battery and the Formal Floor

Protocol	Checks	Result	Lifts / anchors
LTP1 residual flow	C1–C3	[finite_diagnostic] PASS 3/3	L-04, L-06, L-07/08
LTP2 sorites threshold	C1–C3	[finite_diagnostic] PASS 3/3	L-10 (sorites half)
LTP3 bounded quantification	A–B	[finite_diagnostic] PASS 2/2	L-05
LTP4 rank deficiency	C1–C3	[finite_diagnostic] PASS 3/3	L-11 (as identifiability)
`UPL_Sorites.v`	T1–T4	[Th_coqc] (3× Closed)	Thm. 8.1
LTP5 liar loop	—	planned	L-10 (liar half)
LTP6 AGM postulate check	—	planned	Ch. 21 bridge
LTP7 classical-limit regime map	—	planned	Ch. 7 quantitative

The *three-file binding* release criterion (imported from the successor system): a node of the logic stream reaches top grade only when the human-readable chapter, the machine-checked script, and the executable diagnostic all pass simultaneously. The first node to achieve full binding is sorites: Chapter 8 ↔ `UPL_Sorites.v` ↔ LTP2.

Chapter 24

Claim Discipline and the Falsifier Battery

Forbidden claims (auto-fail): "this book derived classical logic from physics"; "this book solved the liar / sorites / Gödel" (relocation is not solution — even machine-checked relocation); "the LTPs prove that knowers *are* residual-flow systems" (existence of a reading, not uniqueness); "defeasibility is proven bridge-free"; "ethical content is derived"; "the aesthetics/politics chapter is established"; "this book supersedes the traditions of Part IV".

Falsifier battery. Inherited universal: exhibit any knower with $\varepsilon_{\text{tot}} = 0$. Native: (U1) a finite knower running strictly classical revision-free logic against a live world with $r \equiv 0$ breaks the Defeasibility Theorem; (U2) contentful negation with no distinguishability substrate breaks L-02; (U3) causally symmetric implication breaks L-03; (U4) $V \equiv 0$ under persistent noise in any admissible triple breaks L-08 and fires the universal falsifier; (U5) a well-formed philosophical problem that Ω_{phil} cannot even locate breaks the one-stream principle for philosophy; (U6) a sorites-type paradox provably world-side breaks L-10.

Scoring of any output claiming this book's results: PASS / PARTIAL / FAIL / AUTO-FAIL; every tag change requires a regression-log entry — no silent lifts.

Chapter 25

Open Problems, Exercises, and the Road to Version 2

Open problems (each with its stance — never bare): O-1 uniqueness of B1 (stance: expect negative settlement — implementation is plural); O-2 the Gödel bridge (stance: expect a mapping through the null space of a self-referential grammar; not done); O-3 AGM representation theorems for residual flow; O-4 the liar formalization; O-5 a quantitative regime map for classical validity; O-6 aesthetics/politics derivation chains; O-7 the formal comparison with discourse ethics.

The road to Version 2: an audit-first curriculum closure. Before v2.0, a philosophy curriculum audit of ∼60–80 canonical problems (Gettier, induction, reference, personal identity, modality, meaning, . . .) will be tracked row by row as DIRECT / MAPPABLE / NEW CLOSURE / Open — so that the next edition claims its coverage with checkable numbers, not impressions.

Exercises.

1. Run LTP1 with a non-identity W. Which philosophical conclusions of Part II depend on the choice of W?
2. Write the grammar A of the syllogism Barbara and show that its validity is a property of the null space.
3. Choose a paradox this book does not treat (Newcomb; the Ship of Theseus) and run the full seven-step Ω_{phil}, including the falsifier of your own answer.
4. Find one claim in this book whose tier tag is too high, and write the argument that lowers it. If you succeed, the book worked as designed.

Closing. A world-class textbook is not one that answers every question, but one that knows the coordinates of its every answer, the price of every bridge it crosses, and bares its own throat to the sharpest knife: exhibit a knower with $\varepsilon_{\text{tot}} = 0$, and this whole book collapses — honestly.

Appendix A

Logic Test Protocols (Executable Source)

A.1 LTP1 — logic as residual flow

```
"""
LTP1 -- Logic as Residual Flow (Universal Philosophy & Logic, canon v0.2)
Reuses the MVS engine from yaoharee-proposal (Aeps=delta, V=1/2r^TWr, eps=-LambdaA^TW r)
but reads it at the LOGIC tau_c:

  A  = logical grammar (relational Jacobian; A2-admissible transport)
  eps  = belief-state deviation of a finite knower
  delta  = world-resistance readout (theta via record operator; noisy, time-varying)
  r  = Aeps-delta  = residual = "incoherence of the belief state w.r.t. evidence"

Three checks (pass criteria stated up front, MVS-style):

  C1 CONVERGENCE   : constant delta => V(t) monotone v, r -> r_min (inference = Lyapunov
    descent)
  C2 DEFEASIBILITY : delta changes at t_s (new evidence) => a previously "concluded"
    proposition
                     REVISES (non-monotonic revision) and V re-descends (ISS tracking)
  C3 IRREDUCIBLE eps : noisy delta(t) (eta>0 floor) => V never reaches 0; time-averaged
    residual
                     bounded AWAY from 0  (operational face of eps_tot > 0, Theorem step 24)

If C1 fails  -> L-04 (inference as path) breaks.
If C2 fails  -> L-06 (defeasibility theorem) breaks.
If C3 gives V->0 exactly under noise -> universal falsifier fires (eps_tot=0).
"""

import numpy as np

rng = np.random.default_rng(42)

# --- grammar: 4 propositions, 3 relational constraints (rank-deficient on purpose:
#     a finite knower's grammar underdetermines the world -> A4 accessibility) ---
A = np.array([
    [1.0, -1.0,  0.0,  0.0],    # b1 - b2 should match evidence e1
    [0.0,  1.0, -1.0,  0.0],    # b2 - b3 should match evidence e2
    [0.0,  0.0,  1.0, -1.0],    # b3 - b4 should match evidence e3
])
m, n = A.shape
W   = np.eye(m)          # A5 admissibility: positive metric
Lam = 0.8 * np.eye(n)    # transport rate
```

```
def V_of(r):  return 0.5 * r @ W @ r

def run(delta_fn, T=60.0, dt=0.01, eps0=None):
    eps = np.zeros(n) if eps0 is None else eps0.copy()
    ts, Vs, epss = [], [], []
    steps = int(T/dt)
    for k in range(steps):
        t = k*dt
        d = delta_fn(t)
        r = A @ eps - d
        eps = eps - dt * (Lam @ (A.T @ W @ r))
        ts.append(t); Vs.append(V_of(A@eps - d)); epss.append(eps.copy())
    return np.array(ts), np.array(Vs), np.array(epss)

report = []

# ---------- C1: constant evidence -> inference converges ----------
d0 = np.array([1.0, 0.5, -0.3])
ts, Vs, epss = run(lambda t: d0)
mono_viol = int(np.sum(np.diff(Vs) > 1e-12))
c1_pass = (Vs[-1] < 1e-8) and (mono_viol == 0)
report.append(("C1␣convergence␣(inference␣=␣Lyapunov␣descent)",
               c1_pass, f"V(0)={Vs[0]:.4f}␣->␣V(end)={Vs[-1]:.2e};␣monotonicity␣violations={
    mono_viol}"))

# ---------- C2: evidence flips mid-run -> conclusion revises ----------
d1 = np.array([-1.0, 0.5, -0.3])          # e1 flips sign at t_s = 30
def delta_switch(t): return d0 if t < 30.0 else d1
ts, Vs, epss = run(delta_switch)
i_pre  = int(29.9/0.01)                    # settled conclusion before switch
i_post = -1                                # settled conclusion after re-descent
concl_pre  = epss[i_pre,0]  - epss[i_pre,1]    # the "b1-b2" judgment
concl_post = epss[i_post,0] - epss[i_post,1]
jump = Vs[int(30.0/0.01)+1]                # V spikes when world resists
c2_pass = (abs(concl_pre - 1.0) < 1e-3) and (abs(concl_post + 1.0) < 1e-3) and (jump > 0.1)
    and (Vs[-1] < 1e-6)
report.append(("C2␣defeasibility␣(non-monotonic␣revision␣under␣new␣resistance)",
               c2_pass, f"judgment(b1-b2):␣{concl_pre:+.3f}␣->␣{concl_post:+.3f};␣V␣spike␣at
    ␣switch={jump:.3f};␣V(end)={Vs[-1]:.2e}"))

# ---------- C3: noisy world -> residual floor never zero ----------
def delta_noisy(t):
    return d0 + 0.15*np.sin(3.0*t)*np.array([1,0,0]) + 0.05*rng.standard_normal(m)
ts, Vs, epss = run(delta_noisy, T=120.0)
V_tail = Vs[len(Vs)//2:]                   # discard transient
floor  = V_tail.mean()
c3_pass = floor > 1e-4                     # bounded away from zero
report.append(("C3␣irreducible␣residual␣(operational␣eps_tot␣>␣0)",
               c3_pass, f"time-averaged␣V␣over␣tail␣=␣{floor:.5f}␣(must␣be␣>␣0);␣min␣V␣in␣
    tail␣=␣{V_tail.min():.5f}"))

print("LTP1␣--␣Logic␣as␣Residual␣Flow␣␣(engine:␣yaoharee-proposal␣MVS␣/␣Aeps=delta)")
print("="*78)
ok = True
for name, p, detail in report:
    ok &= p
    print(f"[{'PASS'␣if␣p␣else␣'FAIL'}]␣{name}\n␣␣␣␣␣␣␣{detail}")
print("="*78)
print("SUITE:", "PASS␣3/3␣--␣L-04,␣L-06,␣step-24␣operational␣anchors␣delivered␣(T_op)" if ok
     else "FAIL")
```

A.2 LTP2–LTP4 — battery

```
"""
LTP2 + LTP3 + LTP4 -- Universal Philosophy & Logic (UPL) test battery, release 2
Engine: same residual-flow core as LTP1 (Aeps=delta, V=1/2r^TWr, eps=-LambdaA^TW r)

LTP2  sorites_as_threshold       -> would lift L-10 [Wf]->[Dr]
LTP3  bounded_quantification     -> would lift L-05 [Dr]->[Dr-anchored]
LTP4  rank_deficiency_goedel     -> would lift L-11 [Cj]->[Dr] (structural analogy only)
"""

import numpy as np
rng = np.random.default_rng(7)
results = []

def add(name, ok, detail): results.append((name, ok, detail))

# ===================================================================
# LTP2 -- SORITES AS RETENTION-THRESHOLD ARTIFACT
# World-side: graded quantity g(n) (e.g. "heapness" vs grains n), smooth.
# Knower-side readout: binary predicate H(n) = [g(n) >= Pi].
# Claims to verify:
#  C1: graded level obeys TOLERANCE: |g(n)-g(n-1)| < tol for all n (no jump in the world)
#  C2: binary readout has EXACTLY ONE discontinuity, located at Pi-crossing
#      -> the "paradoxical step" exists only in the projection
#  C3: moving Pi moves the flip point but leaves g(n) untouched
#      -> the paradox lives in the readout operator, not in the world
# ===================================================================
n_grid = np.arange(1, 10001)
g = 1.0 - np.exp(-n_grid / 1500.0)          # smooth graded "heapness"
tol = np.max(np.abs(np.diff(g)))
c1 = tol < 1e-3                              # world-side tolerance holds

def flips(Pi):
    H = (g >= Pi).astype(int)
    d = np.diff(H)
    return np.flatnonzero(d != 0)

f_05 = flips(0.5); f_08 = flips(0.8)
c2 = (len(f_05) == 1)
c3 = (len(f_08) == 1) and (f_08[0] != f_05[0]) and np.allclose(g, 1.0 - np.exp(-n_grid
    /1500.0))
add("LTP2-C1␣world-side␣tolerance␣(graded␣g␣smooth)", c1, f"max␣step␣|Deltag|={tol:.2e}␣<␣1e
    -3")
add("LTP2-C2␣binary␣readout␣has␣exactly␣ONE␣forced␣jump", c2, f"flip␣at␣n={f_05[0]+1}␣for␣Pi
    =0.5␣(count={len(f_05)})")
add("LTP2-C3␣paradox␣tracks␣Pi,␣not␣the␣world", c3, f"Pi=0.5->flip␣n={f_05[0]+1};␣Pi=0.8->
    flip␣n={f_08[0]+1};␣g␣unchanged")
# Reading: 'tolerance principle' (heap(n)->heap(n-1)) is TRUE of g, FALSE of H exactly
# at one Pi-determined point => sorites = projection artifact of retention threshold Pi. [Dr
    ]

# ===================================================================
# LTP3 -- BOUNDED QUANTIFICATION (inference over S_A  proper-subset-of  Delta)
# Full grammar: 3 constraints over 4 propositions (as LTP1).
# Salient sub-grammar: knower attends only to constraints {1,2} (drops row 3).
# Case A: dropped constraint NOT load-bearing for query q=(b1-b2)  -> bounded ~= full
# Case B: query q'=(b3-b4) DEPENDS on dropped row                  -> bounded diverges
# Pass: divergence occurs IFF the discarded difference is load-bearing.
# ===================================================================
A_full = np.array([[1.,-1.,0.,0.],[0.,1.,-1.,0.],[0.,0.,1.,-1.]])
```

```
A_sal  = A_full[:2]                            # salience-selected grammar
d_full = np.array([1.0, 0.5, -0.3])
d_sal  = d_full[:2]

def settle(A, d, T=80.0, dt=0.01):
    n = A.shape[1]; eps = np.zeros(n); W = np.eye(A.shape[0]); Lam = 0.8*np.eye(n)
    for k in range(int(T/dt)):
        r = A @ eps - d
        eps = eps - dt * (Lam @ (A.T @ W @ r))
    return eps

e_full = settle(A_full, d_full)
e_sal  = settle(A_sal,  d_sal)
qA_full, qA_sal = e_full[0]-e_full[1], e_sal[0]-e_sal[1]     # b1-b2
qB_full, qB_sal = e_full[2]-e_full[3], e_sal[2]-e_sal[3]     # b3-b4
caseA_ok = abs(qA_full - qA_sal) < 1e-3                       # agree when not load-bearing
caseB_div = abs(qB_full - qB_sal) > 0.25                      # diverge when load-bearing
add("LTP3-A␣bounded␣inference␣~=␣full␣when␣discard␣is␣idle", caseA_ok,
    f"(b1-b2):␣full={qA_full:+.3f}␣vs␣bounded={qA_sal:+.3f}")
add("LTP3-B␣bounded␣inference␣diverges␣when␣discard␣is␣load-bearing", caseB_div,
    f"(b3-b4):␣full={qB_full:+.3f}␣vs␣bounded={qB_sal:+.3f}␣(bounded␣knower␣has␣NO␣access␣to
    ␣this␣difference)")
# Reading: 'forall ' of a finite knower quantifies over S_A; validity is relative to whether
# discarded differences are load-bearing for the query -- exactly L-05. [Dr-anchored]

# ======================================================================
# LTP4 -- RANK DEFICIENCY => TRUE-BUT-UNREACHABLE (Goedel-flavored, STRUCTURAL ANALOGY ONLY)
# A_full above has rank 3 < n=4: null(A) = span(1,1,1,1).
# Two distinct world-configurations theta1 != theta2 differing along the null space
# produce IDENTICAL records delta = Atheta, hence identical residual dynamics forever.
# Checks:
#  C1: theta1 != theta2 (a real difference exists)
#  C2: Atheta1 == Atheta2 exactly (records identical) => no admissible inference separates
    them
#  C3: Fisher matrix J^T J is singular (identifiability statement, RUSC-style)
# ======================================================================
theta1 = np.array([0.7, -0.3, 0.2, 0.5])
theta2 = theta1 + 1.3*np.ones(4)               # shift along null space
c1 = not np.allclose(theta1, theta2)
c2 = np.allclose(A_full@theta1, A_full@theta2, atol=1e-12)
F = A_full.T @ A_full
c3 = np.linalg.matrix_rank(F) < 4 and abs(np.linalg.det(F)) < 1e-10
add("LTP4-C1␣a␣genuine␣world-difference␣exists", c1, f"||theta1-theta2||={np.linalg.norm(
    theta1-theta2):.3f}")
add("LTP4-C2␣records␣are␣identical␣=>␣difference␣inaccessible-admissible", c2,
    "Atheta1␣==␣Atheta2␣exactly;␣every␣inference␣trajectory␣coincides")
add("LTP4-C3␣Fisher␣rank-deficient␣(identifiability,␣not␣mysticism)", c3,
    f"rank(A^TA)={np.linalg.matrix_rank(F)}/4,␣det={np.linalg.det(F):.1e}")
# Reading: 'true but unprovable-from-records' realized as null-space truth. This is a
# STRUCTURAL ANALOGY to incompleteness (self-reference NOT constructed here) -- the honest
# lift is L-11 [Cj]->[Dr] as identifiability reading, with the formal Goedel bridge still
    Open.

print("LTP2-LTP4␣battery")
print("="*84)
ok = True
for nm, p, dt_ in results:
    ok &= p; print(f"[{'PASS'␣if␣p␣else␣'FAIL'}]␣{nm}\n␣␣␣␣␣␣␣␣{dt_}")
print("="*84)
print("SUITE:", ("PASS␣8/8␣--␣lifts:␣L-10->[Dr],␣L-05->[Dr-anchored],␣L-11->[Dr␣as␣
    identifiability;␣"
                 "Goedel␣formal␣bridge␣remains␣Open]") if ok else "FAIL")
```

Appendix B

The Formal Floor (Coq)

Compile with `coqc code/UPL_Sorites.v` (Coq 8.18); expected: exit code 0 and three times *Closed under the global context.*

```
(* ==========================================================================
   UPL_Sorites.v -- UPL brick 1 (target tier: Th_coqc, axiom-free over nat)
   Machine-checked core of LTP2 / book chapter 8:
     world-side: a MONOTONE graded quantity g : nat -> nat
     knower-side: binary threshold readout  H(n) := (Pi <=? g n)
   Theorems:
     T1 readout_monotone          : the readout never flips back
     T2 flip_unique               : at most ONE flip point exists
     T3 tolerance_violation_iff_flip : the 'sorites step' (H n <> H (S n))
                                       happens EXACTLY at the flip point
     T4 threshold_order           : a higher threshold can only delay truth
   Reading: the paradoxical step is a property of the Pi-READOUT, provably not
   of the graded world-side g. (Discrete core; no Reals, no classical axioms.)
   ========================================================================== *)

Require Import Arith Lia.

Section Sorites.
  Variable g  : nat -> nat.                      (* world-side graded quantity *)
  Hypothesis g_mono : forall m n, m <= n -> g m <= g n.
  Variable Pi : nat.                             (* retention threshold (RAR: Pi) *)

  Definition H (n : nat) : bool := Nat.leb Pi (g n).

  (* T1 *)
  Lemma readout_monotone : forall m n, m <= n -> H m = true -> H n = true.
  Proof.
    intros m n Hmn Hm. unfold H in *.
    apply Nat.leb_le. apply Nat.leb_le in Hm.
    eapply Nat.le_trans; [exact Hm | apply g_mono; exact Hmn].
  Qed.

  Definition flip_at (a : nat) : Prop := H a = false /\ H (S a) = true.

  (* T2 -- the sorites 'cut' exists at most once *)
  Theorem flip_unique : forall a b, flip_at a -> flip_at b -> a = b.
  Proof.
    intros a b [Ha1 Ha2] [Hb1 Hb2].
    destruct (Nat.lt_trichotomy a b) as [Hlt | [Heq | Hgt]]; auto.
    - exfalso.
      assert (Hb : H b = true).
      { apply (readout_monotone (S a) b); [lia | exact Ha2]. }
      congruence.
```

```
    - exfalso.
      assert (Ha : H a = true).
      { apply (readout_monotone (S b) a); [lia | exact Hb2]. }
      congruence.
  Qed.

  (* T3 -- the 'paradoxical step' is located EXACTLY at the flip *)
  Theorem tolerance_violation_iff_flip :
    forall n, H n <> H (S n) <-> flip_at n.
  Proof.
    intro n. unfold flip_at.
    destruct (H n) eqn:E1; destruct (H (S n)) eqn:E2.
    - split; [congruence | intros [F _]; congruence].
    - (* true -> false is impossible: readout is monotone *)
      exfalso.
      assert (Hs : H (S n) = true).
      { apply (readout_monotone n (S n)); [lia | exact E1]. }
      congruence.
    - split; [intros _; auto | congruence].
    - split; [congruence | intros [_ F]; congruence].
  Qed.

End Sorites.

Section TwoThresholds.
  Variable g : nat -> nat.
  Variables Pi1 Pi2 : nat.
  Hypothesis HPi : Pi1 <= Pi2.

  (* T4 -- the flip point tracks the threshold ordering (LTP2-C3, proof-grade) *)
  Theorem threshold_order :
    forall n, Nat.leb Pi2 (g n) = true -> Nat.leb Pi1 (g n) = true.
  Proof.
    intros n Hn. apply Nat.leb_le. apply Nat.leb_le in Hn. lia.
  Qed.
End TwoThresholds.

(* axiom audit -- must print 'Closed under the global context' for Th_coqc *)
Print Assumptions flip_unique.
Print Assumptions tolerance_violation_iff_flip.
Print Assumptions threshold_order.
```

Appendix C

Claims Register

One falsifiable statement per row; the check command reproduces the quoted result on this bundle.

Claim	Check	Expected
C1. Inference-as-descent converges monotonically under constant evidence	`python3 code/LTP1....py`	C1 PASS; $V \to$ 8.2e−26, 0 violations
C2. A settled conclusion revises under an evidence flip	same	C2 PASS; $+1.000 \to -1.000$
C3. Residual floor never reaches zero under noise	same	C3 PASS; $\bar{V} = 0.00795 > 0$
C4. Sorites: world-side tolerance holds; readout has exactly one jump; jump tracks Π	`python3 code/LTP2_3_4....py`	LTP2 3/3 PASS
C5. Bounded inference diverges iff the discarded difference is load-bearing	same	LTP3 2/2 PASS
C6. Distinct worlds with identical records exist; Fisher singular	same	LTP4 3/3 PASS
C7. Sorites core is machine-checked axiom-free (monotone scope)	`coqc code/UPL_Sorites.v`	exit 0; 3× Closed

We do NOT claim: that any [finite_diagnostic] result is a proof; that the Coq scope extends beyond monotone g; that B1 is unique; that Gödel, the liar, or first-order ethics are solved; machine-independence of any timing.

Appendix D

Arrow Ledger, Borrowed-vs-Derived, and Retractions

Arrow ledger (classes: DEFINITIONAL / FORCED-given-X / CHOSEN / REFUTED / STILL-OPEN): L-01 definitional; L-02 definitional-given-P1 (queued as the next Coq brick); L-03 forced-given-graph-asymmetry; L-04 chosen reading with [finite_diagnostic] support — alternatives (argumentation semantics, sequent search) not excluded; L-06 forced-given-B1, B1 itself chosen; L-07 chosen weld (reclassified down — see R3); L-09 chosen, well-motivated; L-10 forced-given-$\{$monotone g, threshold readout$\}$, machine-checked in scope; L-11 chosen analogy with [finite_diagnostic] witness, Gödel bridge still open; B1 uniqueness still open, adjudication campaign queued (a negative settlement is also a result).

Borrowed (standard material re-read, not created): Lyapunov and ISS theory; the AGM postulates (benchmark only); Fisher identifiability; every historical doctrine of Part IV. **Derived or forced here:** sorites flip-uniqueness under threshold readout ([Th_coqc], scoped); divergence-iff-load- bearing-discard ([finite_diagnostic]); the null-space inaccessibility construction ([finite_diagnostic]).

Retractions. R1: the earlier word "anchor" was ambiguous between proof and numeric; all LTP results are [finite_diagnostic]. R2: any statement of the L-11 lift omitting "as identifiability reading only" overclaims. R3: L-07 was presented with derivational flavour; it is a chosen weld — the reading is the contribution, the mathematics is borrowed.

Appendix E

Glossary

$\theta(E)$ latent world-truth; never directly accessed.

M_A, **record** the knower's model/output; $M_A = K_A\theta + \eta \neq \theta$.

$\varepsilon_{\text{tot}} > 0$ the irreducible error floor; the universal falsifier lives here.

τ_c causal memory time; the only thing that changes across scales.

readout / non-readout what a finite agency can actually read, versus idealizations (the continuum, actual infinity) injected by I1–I4.

retained distinguishability the razor: reality is distinguishability that keeps its effect under admissible transport.

three-file binding release criterion: chapter $\leftrightarrow$ Coq $\leftrightarrow$ executable diagnostic must pass together.

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